

AI & Automation Assignment (Black Friday Scenarios)

Theme 02: Supply Chain Perspective – Inventory Volatility

Within each group, every participant must submit their own solution

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Scenario: The company's supply chain faces **15x normal demand variability** during Black Friday. Real-time tracking of **200,000+ SKUs** across warehouses and last-mile hubs is critical. Manual intervention delays result in **15–20% order fulfillment errors** and increased returns. AI-powered predictive demand modeling and automation of replenishment must be integrated into the architecture.

Problem Statement: How can AI-based predictive analytics and workflow automation ensure seamless inventory optimization, reduce stockouts, and enable proactive rerouting under high volatility?

Assignment Instructions (Individual Work)

- **Objective:** Create an **Architecture Document** (using Arc42 template as reference) for your assigned scenario.
- **Steps:**
 1. Define the business challenge and context.
 2. Document functional & non-functional requirements (include NFRs, Metrics & KPIs).
 3. Propose architecture building blocks and interactions.
 4. Highlight where AI & automation augment the architecture (e.g., fraud detection, predictive analytics, infrastructure orchestration).
 5. Address risks, compliance, and trade-offs.
 6. Ensure traceability of design choices through **Architecture Decision Records (ADRs)**.

Submission: 4–6-page architecture document (Arc42-aligned) with diagrams, NFR table, and AI/automation integration highlights.

Best Practices & Frameworks to Discuss

- **Frameworks:**
 - **Arc42** for documentation structure.
 - **TOGAF ADM** for contextual alignment.
 - **ISO/IEC 25010** for NFR quality attributes.
 - **NIST AI RMF** for AI trust, fairness, and compliance.
 - **MLOps frameworks** for operationalizing AI in architecture.
- **Best Practices:**
 - Always document **AI/automation boundaries** (what's manual vs. automated).
 - Embed **metrics-driven design** (latency, accuracy, cost efficiency).
 - Ensure **explainability & traceability** for AI decisions (critical for BFSI & Insurance).
 - Design **resilience-first** (assume AI pipelines may fail, build graceful fallbacks).
 - Use **ADR rigor** to justify automation vs. manual interventions.